UNISA | college of science, engineering and technology

UNIVERSITY EXAMINATIONS

JANURAY/FEBRUARY 2025

COS1511

Introduction to Programming

Welcome to the COS1511 examination.

Date: 24 January 2025
Time: 13:45
Hours: 2

Examiner name: Ms. Promise Mvelase
Internal moderator name: Ms. Mokateko Buthelezi

This paper consists of 9 pages.

Total marks: 70

Number of pages: 9
<u>Instructions:</u>

- The use of non-programmable calculators is permissible.
- This is a closed book examination.
- The IRIS invigilation tool must be used.
- Answer all the questions on a Word document/by hand and convert to PDF.
- Number your answers and label your rough work clearly.
- Marks are awarded for part of an answer, so do whatever you are able to in each question.
- Students are provided 30 minutes to submit their answer scripts after the official examination time. Students who experience technical challenges should report the challenges to the SCSC on 080 000 1870 or their College exam support centres (refer to the Get help during the examinations by contacting the Student Communication Service Centre [unisa.ac.za]) within 30 minutes. Queries received after 30 minutes of the official assessment duration time will not be responded to. Submissions made after the official assessment time will be rejected according to the examination regulations and will not be marked. Only communication received from your myLife account will be considered.

Additional student instructions

1. Students must upload their answer scripts in a single PDF file (answer scripts must not be password protected or uploaded as “read only” files)
2. Incorrect file format and uncollated answer scripts will not be considered.
3. NO emailed scripts will be accepted.
4. Students are advised to preview submissions (answer scripts) to ensure legibility and that the correct answer script file has been uploaded.
5. Incorrect answer scripts and/or submissions made on unofficial examinations platforms (including the invigilator cell phone application) will not be marked, and no opportunity will be granted for resubmission. Only the last answer file, uploaded on the official examination platform within the stipulated submission duration period will be marked.

6. Mark awarded for incomplete submission will be the student's final mark. No opportunity for resubmission will be granted.
7. Mark awarded for illegible scanned submission will be the student's final mark. No opportunity for resubmission will be granted.
8. Submissions will only be accepted from registered student accounts.
9. Students who have not utilised the proctoring tool will be deemed to have transgressed Unisa's examination rules and will have their marks withheld. If a student is found to have been outside the proctoring tool for a total of 10 minutes during their examination session, they will be considered to have violated Unisa's examination rules and their marks will be withheld. For examinations which use the IRIS invigilator system, IRIS must be recording throughout the duration of the examination until the submission of the examinations scripts.
10. Students have 48 hours from the date and time of their examination to upload their invigilator results from IRIS. Failure to do so will result in students deemed not to have utilized the proctoring tools.
11. Students suspected of dishonest conduct during the examinations will be subjected to disciplinary processes.
12. Students may not communicate with any other person or request assistance from any other person during their examinations.
13. Plagiarism is a violation of academic integrity and students who plagiarise, copy from published work or Artificial Intelligence Software (eg ChatGPT) or online sources (eg course material), will be in violation of the Policy on Academic Integrity and the Student Disciplinary Code and may be referred to a disciplinary hearing. Unisa has a zero tolerance for plagiarism and/or any other forms of academic dishonesty.
14. Listening to audio (music) and making use of audio-to-text software is strictly prohibited during your examination session unless such usage of the software is related to a student's assistive device which has been so declared. Failure to do so will be a transgression of Unisa's examination rules and the student's marks will be withheld.
15. Non-adherence to the processes for uploading assessment responses will not qualify the student for any special concessions or future assessments.
16. Queries that are beyond Unisa's control include the following:
    a. Personal network or service provider issues;
    b. Load shedding/limited space on personal computer;
    c. Crashed computer;
    d. Non-functioning cameras or web cameras;
    e. Using work computers that block access to the myExams site (employer firewall challenges);
    f. Unlicensed software (e.g. license expires during exams).

    Postgraduate students experiencing the above challenges are advised to apply for an aegrotat and submit supporting evidence within ten days of the examination session. Students will not be able to apply for an aegrotat for a third examination opportunity.
    Postgraduate/undergraduate students experiencing the above challenges in their second examination opportunity will have to reregister for the affected module.
17. Students experiencing network or load shedding challenges are advised to apply together with supporting evidence for an Aegrotat within 3 days of the examination session.

Question 1 **[20]**

Question 1 presents multiple choice questions and several possible answers as options for you to choose from. Choose the correct option and write only the number of the correct answer.

1.What is the output of the following code fragment? **(2)**

```
int x = 0;
while( x < 8)
cout << x << " ";
x ++;
cout << x << endl;
```

a) 0
b) infinite loop
c) 1 2 3 4 5 6 7 8
d) 0 1 2 3 4 5 6 7 8

2.What is the value of `x` after the following statements have executed? **(2)**

```
int x, y, z; y = 10;
z = 3;
x = y * z - 3;
```

a) 0
b) 3
c) 10
d) 27

3.What is the value of `x` after the following statements have executed? **(2)**

```
int x;
x = x + 15;
```

a) 0
b) 14
c) 15
d) not defined

4.What is the output of the following code? **(2)**

```
cout << "Where is the \\" << endl;
```

a) Where is the \
b) 2.Where is the
c) 3.Nothing, it is a syntax error
d) 4.Where is the \ endl

5.What is the value of x after the following statements have executed? **(2)**

```
int x;
x = 19 % 5;
```

a) 4.0
b) 2. 4
c) 3. 3.8
d) 4. 8

6.Given the following code fragment and the input value of 5, what output will be generated? **(2)**

```
float tax; float total;

cout << "Enter the cost of the item\n"; cin  >> total;
if ( total >= 5.0)
{
  tax = 0.14;
  cout << total + (total * tax) << endl;
}
else
{
  cout << total << endl;
}
```

a) 5
b) 5.7
c) 11
d) 11.4

7.Given the following code fragment and the input value of 3.0, what output is generated? **(2)**

```
float tax; float total;
cout << "Enter the cost of the item\n"; cin  >> total;
if (total > 3.0)
{


}
else
{

}


tax = 0.10;
cout << total + (total * tax) << endl;
```

a) 3.0
b) 2.0
c) 3.3
d) 2.3

8.If `x` has the value of 3, `y` has the value of -2, and `w` has the value of 10, is the following condition true or false? **(2)**

```
if( x < 2 && w < y)
```

a) true && true
true

b) false && false
false

c) false && false
true

d) true && false
false

9.Given the following code fragment, and an input value of 8, what is the output that is generated? **(2)**

```
int x;
cout << "Enter a value\n";
cin >> x;

if (x = 0)
{
  cout << "x has no value\n";
}
else
{
  cout << "x has a value\n";
}
```

a) x has no value
b) x is 8
c) x has a value
d) unable to determine

10.Which of the following data types cannot be used in a switch controlling expression? **(2)**

a) int
b) enum
c) array
d) char

QUESTION 2 [20]

Question 2a (1 mark each x 6)

State what output, if any, results from each of the following statements. Write only the solution.

	CODE	OUTPUT

Example	`for (int i = 0; i < 10; i++)`
	`cout << i;`
	`cout << endl;`
		0123456789

a.	`for (int i = 1; i <= 1; i++)`
	`cout << "*";`
	`cout << endl;`	

b.	`for (int i = 1; i <= 10; i++)`
	`cout << i << "     ";`
	`cout << endl;`	

c.	`for (int i = 2; i >= 2; i++)`
	`cout << "*";`
	`cout << endl;`	

d.	`for (int = 12; i >= 9; i--)`
	`cout << "*";`
	`cout << endl;`	

e.	`for (int i = 0; i <= 5; i++)` `cout << "*";`	

	`cout << endl;`	

f.	`for (int i = 1; i <= 5; i++)`
	`cout << "*";`
	`i = i + 1;`
	`cout << endl;`	

Question 2b **(5)**

Suppose we want to find a student that qualifies for an internship. For each student, we input the name, the age of student and the final mark obtained for the examination in a `while` loop. To qualify, the student should be younger than 30 with a final mark of more than 65%. Read in values until a suitable candidate is found. Display appropriate messages, whether successful or not. The variable names are `name`, `age` and `finalMark` respectively. Complete the `while` loop below. You only have to write down the completed `while` loop.

```
string name
cout << "Enter name: ";
cin  >> name;
cout << "Enter age: ";
cin  >> age;
cout << "Enter final mark for exam: ";
      cin  >> finalMark;

//while loop to find a suitable candidate
cout << name << " qualifies for the internship " << endl;
```

Question 2c **(8)**

The program given below gives the output below. Fill in the missing code and write down only the answer for each of the questions 2.c.1 to 2.c.8. Remember to number your answers.

The output:

```
Please enter 10 integers, positive, negative, or zeros.
The numbers you entered are:
2
7
-4
-3
0
7
4
0
-9
-4

There are 6 evens, which includes 2 zeros.
The number of odd numbers is: 4
```

```
#include <iostream>
using namespace std;

const int LIMIT = 2c.1_________
```

```
int main ()
{
    2c.2_________ counter;
    int number;

    int zeros = 0;
    int odds = 0; int
    evens = 0;

    cout << "Please enter " << LIMIT << " integers, "
         << "positive, negative, or zeros." << endl;

    cout << "The numbers you entered are:" << endl;
    for (int counter = 1; counter <= 2c.2.1----- ; counter++)
    {
        cin 2c.3----- number;
          2c.4----- (number % 2)
        {
            case 0:
                  evens++;
                  if (number == 0)
                  zeros++;
              2c.5-----;
            case 1:

              2c.6-----:
              2c.7-----;
        }
    }
    cout << endl;

    cout << "There are " << 2c.8----- << " evens, "
         << "which includes " << zeros << " zeros."
         << endl;
    cout << "The number of odd numbers is: " << odds
         << endl;

    return 0;
}
```

QUESTION 3 [30]

In this question, we describe the problem and then you have to decide yourself how you are going to tackle it.

Question 3a (10)

The cost of renting a room at a hotel is R900 per night. For special occasions, such as a wedding or conference, the hotel offers a special discount as follows:

- if the number of rooms booked is at least 10, the discount is 10%;
- if the number of rooms booked is at least 20, the discount is 20%;
- if the number of rooms booked is greater or equal to 30, the discount is 30%;
- In addition, if rooms are booked for at least three days, there is an additional 5% discount.

Write a program that prompts the user to enter the cost of renting one room, the number of rooms booked, the number of days the rooms are booked and these laws tax (as a percent).

Display the output as follows:

```
Please enter the following:
            cost per room: 1000 sales tax
                 per room: 10
      the number of rooms: 35
           number of days: 2

The total cost for one room is R1000
The discount per room is 30%
The number of rooms booked: 35
The total cost of the rooms are R: 70000
The sales tax paid is : 10%
The total cost per booking is R77000
```

Question 3b (10)

Four experiments are performed, each consisting of five test results. The results for each experiment are given in the following list.

1st experiment results:	23.2	31	16.9	27	25.4
2nd experiment results:	34.8	45.2	27.9	36.8	33.4
3rd experiment results:	19.4	16.8	10.2	20.8	18.9
4th experiment results:	36.9	39	49.2	45.1	42.7

Complete the program below using a nested loop to compute and display the average of the test results for each experiment. Display the average with a precision of two digits after the decimal point.

```
#include <iostream>
using namespace std;
int main()
{
    3b. 1-----result, average;

    for (int exp = 0; 3b.2-----)
    {
        3b.3----- //initialize total
        cout << "Please enter results for experiment no " << exp + 1
                     << ": " << endl;

        for (int i = 0; 3b.4-----)
        {
            cout << "Result no " << i + 1 << ": ";
            cin >> 3b.5-----;
            3b.6-----;//calculate total
        }
        3b.7----- //calculate average
        //Settings to display the average with a precision of two digits
        after the decimal point
        3b.8-----;
        3b.9-----;
        cout << "Average for experiment no " << exp + 1 << ": "
            << 3b.10----- << endl << endl;  //display average
    }
    return 0;
}
```

Question 3c **(10)**

In this program, you have to use the `switch` statement.
The average life expectancy (in hours) of a lightbulb based on the bulb's wattage is listed in the table below:

Watts	Life expectancy (hours)
25	25000
40	1000
60	1000
75	750
100	750

Write a program that when given a bulb's wattage, displays the average life expectancy.

[Total Marks: 70]